

Proposal Report

Data Science and Machine Learning Applications Using CVE Vulnerability Dataset

1. Introduction

The collected dataset contains cybersecurity vulnerability information obtained from the Open Source Vulnerability (OSV) database. The dataset includes CVE IDs, severity levels, CVSS scores, publication dates, vulnerability descriptions, affected software ecosystems, packages, CWE classifications, and other vulnerability-related information.

After completing the data cleaning process, the dataset can be used for different data science and machine learning tasks. This proposal report presents four different proposed applications that can be implemented using the collected CVE dataset.

Proposal 1: Vulnerability Severity Classification

Objective

The first proposal is to develop a machine learning classification model that predicts the severity level of a vulnerability based on the available vulnerability information.

Work to Be Performed

The cleaned dataset will be analyzed using attributes such as vulnerability description, CVSS-related information, CWE category, affected ecosystem, and package information. The severity category, such as **Critical, High, Medium, or Low**, will be used as the target variable.

Text-based fields such as the vulnerability summary and details can be converted into numerical features using **TF-IDF (Term Frequency-Inverse Document Frequency)**. Other categorical attributes can also be encoded for machine learning.

Proposed Models

The following classification algorithms can be tested:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

The models can be compared using accuracy, precision, recall, F1-score, and a confusion matrix.

Expected Output

The model is expected to predict the severity category of a vulnerability. For example, when vulnerability information is provided, the model may classify it as **High** or **Critical** severity.

The final output will include model performance metrics and a comparison of the tested algorithms.

Expected Benefit

This system could help security analysts prioritize vulnerabilities by automatically assigning an estimated severity category to newly collected vulnerability information.

Proposal 2: CVSS Score Prediction

Objective

The second proposal is to develop a regression model that predicts the numerical **CVSS score** of a vulnerability.

Work to Be Performed

CVSS score will be treated as the target variable. Features such as CWE category, affected ecosystem, package information, vulnerability description, publication information, and other available attributes will be used as input features.

Text descriptions can be converted into numerical representations using TF-IDF, while categorical fields can be encoded before training the regression models.

Proposed Models

The following regression models can be implemented:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor

The models can be evaluated using **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, **Root Mean Squared Error (RMSE)**, and **R² score**.

Expected Output

The trained model will produce an estimated CVSS score for a vulnerability. For example, a vulnerability may receive a predicted score such as **7.8**.

The actual CVSS score and predicted score can then be compared to determine the accuracy of the prediction.

Expected Benefit

CVSS score prediction could provide an initial estimate of vulnerability severity when complete scoring information is not immediately available.

Proposal 3: Vulnerability Trend and Pattern Analysis

Objective

The third proposal is to analyze historical vulnerability data to identify patterns and trends in cybersecurity vulnerabilities.

Work to Be Performed

The cleaned dataset will be analyzed based on publication year, severity, CWE category, affected ecosystem, CVSS score, and affected packages.

The analysis will identify:

- Number of vulnerabilities reported each year
- Most common severity levels
- Most frequently occurring CWE categories
- Most affected software ecosystems
- Frequently affected packages
- Average CVSS score by year
- Distribution of vulnerabilities by severity

Proposed Models and Techniques

This proposal will mainly use **Exploratory Data Analysis (EDA)** and statistical techniques rather than a traditional prediction model.

The following techniques can be used:

- Grouping and aggregation
- Correlation analysis
- Time-series trend analysis
- Frequency analysis

- Data visualization
- Clustering using K-Means, if suitable

K-Means clustering can additionally be used to group vulnerabilities based on characteristics such as CVSS score, severity-related features, and other numerical attributes.

Expected Output

The expected outputs include:

- Year-wise vulnerability charts
- Severity distribution charts
- CWE frequency charts
- Ecosystem distribution charts
- CVSS score statistics
- Vulnerability clusters
- Identification of major vulnerability patterns

Expected Benefit

This analysis can help identify historical cybersecurity trends and determine which vulnerability types, software ecosystems, and weakness categories appear most frequently in the collected dataset.

Proposal 4: Vulnerability Description Classification Using NLP

Objective

The fourth proposal is to use **Natural Language Processing (NLP)** to automatically classify vulnerability descriptions according to their CWE categories.

Work to Be Performed

The vulnerability **Summary** and **Details** fields will be treated as text data. The text will be cleaned by removing unnecessary characters, converting text to a consistent format, and removing irrelevant words where appropriate.

The cleaned text will then be converted into numerical features using **TF-IDF** or similar text-vectorization techniques.

The CWE category will be used as the target class. The machine learning model will learn relationships between vulnerability descriptions and their corresponding CWE classifications.

Proposed Models

The following NLP classification models can be tested:

- Naive Bayes
- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest

Model performance will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

Expected Output

When a new vulnerability description is provided, the trained model will predict its most likely CWE category.

For example, based on the vulnerability description, the model could predict categories such as:

CWE-79 – Cross-Site Scripting

or

CWE-89 – SQL Injection

depending on the learned patterns in the dataset.

Expected Benefit

Automated CWE classification could reduce manual analysis effort and help security researchers quickly categorize large numbers of vulnerability descriptions.

5. Overall Expected Outcome

The four proposed applications demonstrate different ways the collected CVE dataset can be used for data science and machine learning:

Proposal	Task	Model/Technique	Expected Output
1	Severity Classification	Random Forest, SVM, Decision Tree	Critical/High/Medium/Low prediction
2	CVSS Prediction	Regression models	Predicted CVSS score
3	Trend & Pattern Analysis	EDA, statistics, K-Means	Trends, patterns, clusters, charts
4	CWE Classification	NLP + ML	Predicted CWE

Proposal	Task	Model/Technique	Expected Output category
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These proposals cover **classification, regression, exploratory analysis, clustering, and NLP-based machine learning** using the same cleaned cybersecurity dataset.

6. Conclusion

The cleaned CVE vulnerability dataset provides several opportunities for practical data science and machine learning applications. The four proposed projects focus on vulnerability severity classification, CVSS score prediction, vulnerability trend analysis, and automatic CWE classification using NLP.

The proposed models will be trained and evaluated using the collected dataset, and their results will be compared using appropriate performance metrics. The final outputs will include predictions, statistical findings, visualizations, and model evaluation results.

These four proposals demonstrate how a web-scraped cybersecurity dataset can be transformed from raw vulnerability information into useful predictive and analytical solutions for cybersecurity.